



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018 – 0483



Agricultural Machinery – Chipping Machine SUKI BS-1

Test Report Number	2018 – 0483
Date of Issuance	August 22, 2019
Valid Until	August 22, 2024
Agricultural Machinery	Chipping Machine
Type	Disc type, Two-blade, Horizontal single shaft, Cart-type
Brand	SUKI
Model	BS-1
Test Requested by	SUKI TRADING CORPORATION Agus, Ibabao, Lapu-Lapu City, Cebu

SUMMARY

Performance Criteria	Requirement as per PAES 222:2005	AMTEC Test
Coefficient of variation of chip thickness, %, maximum	5	NM
Noise level, dB(A), maximum	92.0*	98.2
Quality of output	Clear-cut chipped material	Clear-cut chipped material

Note: *Allowable noise level for six (6) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines, 1983.

NM- Not Measured

OBSERVATIONS

1. Operator's Manual	Not Provided
2. Previous Valid Test Reports of Similar Brand and Model	None
3. Manufacturer	SUKI TRADING CORPORATION
4. Prime Mover	5.22 kW KUBOTA RK70S Diesel Engine
5. Number of Operator/s	One Feeding Operator
6. Transporting the Machine	The machine has wheels for transport
7. Adjustments/Repairing Parts	Not observed
8. Cleaning Parts	Parts were accessible for cleaning.
9. Safety	Belt-pulley transmission system was not provided with cover.
10. Other Remarks	No breakdown occurred.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	March 21, 2018
Location of Test	Agus, Ibabao, Lapu-Lapu City, Cebu
Items Tested	Speed of components, Chipping capacity, Chipping efficiency, Fuel consumption, and Noise level

METHODS OF TEST

The chipper was tested based on the PNS/PAES 223:2004 Agricultural Machinery – Chipping Machine – Methods of Test.

DESCRIPTION OF THE MACHINE



Figure 1. Parts of the SUKI BS-1 Chipper powered by 5.22 kW KUBOTA RK70S Diesel Engine include the following: (a) Loading hopper, (b) Chipping chamber, (c) Output chute, and (d) Prime mover.

SPECIFICATIONS

Item		Manufacturer's Specifications ^a	AMTEC Verification ^b
1	Main Structure		
1.1	Overall dimensions, mm		
1.1.1	Length	ND	1590
1.1.2	Width	ND	830
1.1.3	Height	ND	1255
1.1.4	Weight without prime mover, kg	ND	NM
2	Loading Hopper		
2.1	Material	ND	Checkered plate 2 mm
2.2	Bottom opening, mm		
2.2.1	Diameter	ND	250
2.2.3	Height from the ground, mm	ND	1170
3	Chipping Assembly		
3.1	Blade cover		
3.1.1	Type	ND	Closed cylinder
3.1.2	Material	ND	B.I. Pipe, Sch. 40
3.1.3	Size, L x D, mm	ND	380 x 280
3.2	Chipping blade		
3.2.1	Material	ND	Tool steel
3.2.2	Dimensions, mm		
3.2.2.1	Length	ND	100
3.2.2.2	Width	ND	60
3.2.2.3	Thickness	ND	10
3.2.4	Number of blades	ND	2
3.2.3	Chipping plate		
3.2.3.1	Diameter, mm	ND	300
3.2.3.2	Thickness, mm	ND	6
3.4	Chipper opening bottom, D, mm	ND	50

Note: ^a The requesting party did not submit machine specifications.

^a Data obtained from actual test and on-site measurements.

ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specifications ^a	AMTEC Verification ^b
4	Output chute		
4.1	Material	ND	M.S. Plate 2 mm
4.2	Dimensions, mm		
4.2.1	Length	ND	180
4.2.2	Width	ND	200
5	Transmission System		
5.1	Type	ND	V-Belt & pulley
5.2	Engine stub pulley	ND	102 mm x 2B
5.3	Chipper shaft pulley	ND	305 mm x 2B x 51 mm
5.4	Belt size	ND	B – 30
6	Main Frame, mm	ND	C-channel (51 x 102 x 51)
7	Prime Mover		
7.1	Brand	ND	KUBOTA
7.2	Model	ND	RK70S
7.3	Manufacturer	ND	ND
7.4	Serial number	ND	ND
7.5	Type	ND	Four-stroke cycle, horizontal single cylinder, diesel engine
7.6	Rated power, kW	ND	5.22
7.7	Rated speed, rpm	ND	2400
7.8	Cooling system	ND	Water-cooled
7.9	Starting system	ND	Manual cranking lever
8	Safety Devices	ND	None
9	Special Features	ND	It can be used as shredder.

Note: ^a The requesting party did not submit machine specifications.

^a Data obtained from actual test and on-site measurements.

ND – No Data

RESULTS

Item		Data	
1 Test Material Conditions			
1.2	Type	Ipil-Ipil stalk	
1.3	Source	Unknown	
1.4	Moisture content, % _{wb}	51.8	
1.5	Dimension, L x D, mm	NM x 37	
2 Performance Test			
2.1	Input weight, kg	22	
2.2	Operating Time, min	1.92	
2.3	Chipping capacity, kg/h	688.7	
2.4	Chipping efficiency, %	100	
2.5	Coefficient of variation, %	NM	
2.6	Speed of components, rpm	Without Load	With Load
2.6.1	Engine	2539	2445
2.6.2	Chipper shaft	2492	2442
2.7	Noise level, dB (A)		
2.7.1	Feeding operator	90.9	98.2
2.7.2	Bagger	93.0	95.0
2.8	Fuel consumed, mL	100	
2.9	Fuel time, min	4.17	
2.10	Fuel consumption, L/h	1.45	

OBSERVATIONS



Figure 2. Quality of chipped materials.

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AUG 22 2019
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